SPE-227680-MS

Shallow Geo-hazard Detection: Implications on Field Development

A. M. Alzahrani, S. N. Awfi, Q. Fu, and O. A. Alsalmi, Saudi Aramco, Dhahran, Saudi Arabia

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

This paper outlines a novel method to improve detection rates of potential shallow formation drilling geo-hazards by identifying geologic anomalies using 3D seismic data. The resulting identification of these anomalies can be utilized to reduce drilling costs and associated lost time from lost circulation zones. The results are compared to the previously used shallow geo-hazard detection method, which was part of a de-risking strategy.

The drilling geo-hazards we aim to identify, comprises of highly porous carbonates in the form of caverns and sinkholes, also known as karsts. Identifying these geo-hazards from seismic data was accomplished by first flattening the seismic volume using our novel flattening algorithm, which requires no horizon interpretation on which to flatten. From that flattened seismic volume, we then extract an in-house developed edge detection attribute with a Sobel filter, which was capable of outlining the geologic anomalies and discontinuities in the data associated with those geo-hazards.

The results from using this method showed a 10% improvement in detecting these geologic anomalies and in greater detail, compared to the previous approach, once correlated with legacy wells that have encountered lost circulation from drilling through these anomalies. Our new method applies a novel flattening algorithm to the input 3D seismic volume. This algorithm is superior to traditional horizon flattening methods, not only because it requires no horizon interpretation, but it also employs the vectorized disorder algorithm (Fu & Al Dossary, 2021) that further enhances the flattening results. The extraction of the edge detection attribute with an enhanced Sobel-based filter displayed a significant improvement over the previously used variance attribute. The variance attribute outputs the variance of the values within a radius from a center point with an optional smoothing operator, whereas the Sobel-based method works with trace correlations instead of values, allowing it to detect only the geological discontinuities of the input seismic data. This new method is a quick and reliable technique that can be used during field appraisal and development well proposals, to identify and avoid occurrences of drilling losses from geo-hazards.

The novelty lies in this method's ability to bypass traditional seismic horizon interpretation, a step that has historically introduced potential errors into geo-hazard detection, and provides a more consistent approach in doing so. The integration of the enhanced edge detection attribute, coupled with the sophisticated and novel flattening algorithm, represents a significant advancement in seismic analysis, offering a novel and robust tool for cutting drilling costs and improving efficiency.

Introduction

The development of oil and gas plays presents a unique challenge encountered during drilling operations, namely the presence of shallow geo-hazards, particularly karst-related subsurface features. Karsts, formed through the prolonged dissolution of carbonate rocks such as limestone and dolomite, present serious operational risks due to their irregular geometry and cavernous, unpredictable nature. Encountering these features during drilling can lead to substantial operational setbacks, including severe mud losses, borehole instability, and well abandonment. Consequently, the accurate and timely detection of these geological hazards before drilling is critical for the safe and economic exploitation of hydrocarbon resources.

Seismic data interpretation plays a crucial role in identifying geo-hazards. Advances in 3D seismic imaging techniques, particularly the use of edge detection attributes, have significantly improved the precision of geo-hazard detection. This study introduces a new approach that combines a geologically optimized, in-house developed Sobel-based edge detection attribute with a seismic volume flattened using a novel, horizon-independent algorithm. The objective is not only to enhance geo-hazard detection accuracy but also to streamline the interpretation workflow, significantly reducing manual intervention, interpretation bias, and associated drilling risks.

Geological Setting

The shallow geo-hazards (karsts) targeted in this study predominantly formed during the Aptian age of the Early Cretaceous period, a time characterized by widespread deposition of carbonate lithologies. These hazards are associated with thick, massive carbonates exhibiting varying degrees of dolomitization, indicative of a shallow marine depositional environment. Dolomitization resulted from the interaction between the carbonate sediments and magnesium-rich fluids.

These dolomite bodies are typically well-defined and distinguishable, characterized by coarse crystalline textures, high porosity, and frequent vug development—all of which contribute to their cavernous nature. They are often surrounded by less porous and less permeable lithologies, such as anhydrite, which can act as barriers to fluid flow and contribute to localized karst development (Fig. 1). Karstification occurs when percolating water dissolves the soluble dolomite bedrock, exploiting pre-existing fractures and gradually enlarging them through sustained chemical weathering (Fig. 2). This process leads to the formation of interconnected drainage systems, voids, and subsurface cavities.

The geological significance of karst-related geo-hazards stems from their structural instability and spatial unpredictability. Over time, distinctive karst topographies begin to form, including dolines (sinkholes), karst canyons, springs, and sinking streams. The presence of such features, along with associated subsurface voids and conduits, can dramatically impact drilling operations by introducing high levels of uncertainty and risk.

The composition of the formation varies spatially, with different types of carbonate facies encountered at different well locations. A common characteristic across these intervals, however, is high porosity—mainly due to the abundance of vugs and fractures, which enhances permeability and often results in elevated rates of penetration (ROP) during drilling. When wells intersect karstified zones, severe operational challenges such as lost circulation, borehole collapse, wellbore instability, and stuck pipe are frequently encountered.

Despite these risks, the formation also holds significant economic value. In some offshore fields, these dolomitized carbonates serve as prolific hydrocarbon reservoirs, with several producing fields and others under development. The dual nature of this formation, as both a reservoir and a geo-hazard zone, underscores the importance of accurate karst detection and characterization during field planning and drilling operations.

Figure 1—Cratered landscape in South Canterbury, New Zealand. (Rights: Lloyd Homer, GNS Science).

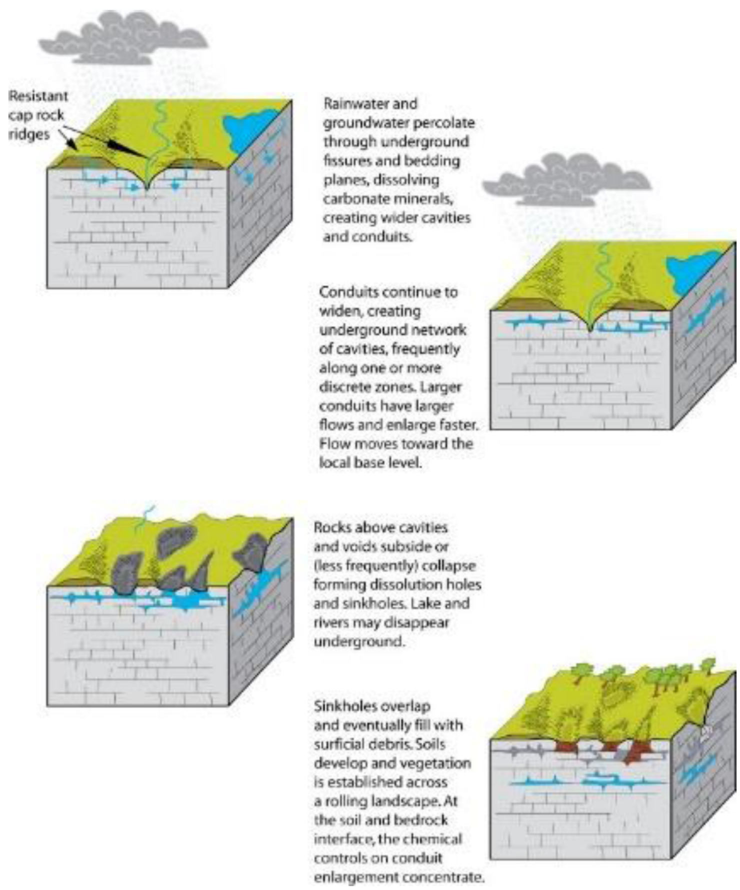

Figure 2—Series of generalized cross-sectional views of the development of a karst landscape. (Illustration by Trista L. Thornberry-Ehrlich, Colorado State University).

Edge Detection Background

Over the last few decades, various 3D seismic attributes have been developed and refined to enhance the interpretability of seismic data, particularly for identifying structural discontinuities associated with faults, fractures, stratigraphic boundaries, and geo-hazards such as karsts. Among these, edge detection attributes stand out due to their ability to highlight abrupt changes in seismic reflection amplitudes, thereby effectively delineating geological discontinuities.

One of the earliest studies in seismic edge detection was conducted by Luo et al. (1996), who introduced two methods: The Difference Method and the Derivative Method. The Difference Method detects edges by evaluating amplitude differences between neighboring seismic traces, effectively capturing abrupt lateral changes. Conversely, the Derivative Method calculates gradients of seismic amplitudes to identify subtle discontinuities, improving detection resolution. Both methods demonstrated enhanced fault delineation compared to traditional seismic interpretation approaches; however, they were prone to noise-related artifacts.

Building on these early efforts, Russell and Ribordy (2014) explored various advanced seismic edge detection algorithms, with a particular focus on derivative-based operators. The Sobel filter, borrowed from digital image processing, computes gradients to highlight edges. While effective in seismic applications, standard Sobel filters faced limitations related to noise sensitivity, particularly at amplitude zero crossings.

Another widely used semblance-based attribute is the variance attribute. As described by Alkhunaini et al. (2022), the variance attribute calculates the variability of seismic amplitudes within a specified neighborhood around each point. Despite its utility, variance attributes have limitations, notably their susceptibility to noise and limited ability to resolve fine-scale karst features.

Current Gaps and Limitations

Despite substantial progress in 3D seismic attribute analysis, several limitations remain. Traditional seismic flattening methods—an essential preprocessing step for edge detection—typically depend on manually interpreted horizons, which introduces bias and subjectivity from the interpreter. This reliance becomes particularly problematic in structurally complex or poorly imaged geological settings, where consistent horizon picking is challenging.

Conventional edge detection methods also face notable challenges. They often struggle with seismic noise, acquisition-related artifacts, and insufficient sensitivity to subtle discontinuities—features that are especially relevant in karst-prone formations. These limitations reduce the reliability of hazard detection and hinder consistent interpretation across varying data quality conditions.

Given these constraints, there is a clear need for advanced, automated workflows that minimize manual intervention, improve interpretive consistency, and enhance the resolution of subtle geological features. The approach proposed in this study—combining a horizon-independent flattening algorithm with an improved Sobel-based edge detection filter—directly addresses these gaps by providing a more robust and scalable solution for identifying shallow geo-hazards.

Methodology

The proposed workflow consists of two main stages: (1) Horizon-independent flattening on a pre-stack time migrated 3D seismic volume and (2) Edge detection using an in-house developed Sobel-based filter. Each stage is designed to overcome limitations commonly encountered in conventional seismic interpretation and is specifically optimized to enhance the detection of shallow geo-hazards, such as karsts.

Horizon-Independent Seismic Flattening

Traditional flattening workflows rely heavily on manually interpreted 3D seismic horizons to restore stratigraphic layers to a reference datum. While effective in well-imaged, structurally simple settings, this approach is time-consuming and highly susceptible to interpreter bias. These limitations become more pronounced in geologically complex or noisy areas, where continuous, clearly defined horizons may not be available.

To address these challenges, the proposed method utilizes a data-driven flattening algorithm that eliminates the need for predefined horizons. Instead, it applies numerical techniques based on the structural geometry of the seismic volume, enabling consistent flattening across varied stratigraphic and structural settings.

To achieve flattening, the structure tensors of the volume are calculated from the numerical gradient of the 3D seismic data cube. Subsequently, eigen decomposition is performed on the structure tensor, and the eigen vector with the largest eigen value is used to get the dips at each point in the 3D volume. The dips are then used to flatten the volume by solving the inversion problem in E.q.1, where s(t, x, y) is the time shift function at each point to flatten the whole 3D cube, and w(x, y, z) is a weight function that assigns greater importance to the regions with less uncertainty (or with higher confidence). In practice, the vectorized disorder algorithm (Fu & Al Dossary, 2021) is a good choice for estimating this weight function. To account for faults or any other discontinuities in the subsurface, constraints can be set to align the points crossing the discontinuities.

$$\begin{bmatrix} w(x,y,z)\left(-\frac{\partial s}{\partial x}-p(x,y,z)\frac{\partial s}{\partial z}\right) \\ w(x,y,z)\left(-\frac{\partial s}{\partial y}-q(x,y,z)\frac{\partial s}{\partial z}\right) \\ \mu\frac{\partial s}{\partial z} \end{bmatrix} \approx \begin{bmatrix} w(x,y,z)p(x,y,z) \\ w(x,y,z)p(x,y,z) \\ 0 \end{bmatrix}, \quad (1)$$

The output of the flattening stage is a geometrically consistent and stratigraphically flattened seismic volume in which subtle lateral variations, such as karst-related discontinuities, become more easily detectable.

Edge Detection Using Sobel-based Filter

With the 3D seismic volume flattened, an edge detection attribute is computed to highlight structural discontinuities. In this workflow, a modified 3D Sobel-based filter is applied, specifically adapted to improve edge clarity and reduce noise sensitivity in geological datasets.

The in-house developed Sobel-based filter is an adaptation of the standard 2D image Sobel filter, extended with several modifications to optimize its performance on seismic data. The first enhancement involves operating on trace correlations instead of values, which is crucial for detecting only the geologically meaningful discontinuities. Another enhancement is the utilization of the complex trace (the real trace plus its Hilbert transform) in computation to reduce noise at zero crossing. The algorithm is also enhanced with a dip guiding feature that enables edge detection only along the dip of the reflectors, thereby avoiding dip-related artifacts. Together, these enhancements significantly improve the quality of the outputs when applied to seismic data. This enhanced Sobel method is a significant improvement over the variance attribute. The variance attribute outputs the variance of the values within a specified radius from a center point, with optional smoothing operations. The variance is usually prone to noise at zero crossings and at layer boundaries, which reduces its quality.

The result is a 3D seismic attribute volume in which discontinuities—such as karst boundaries, collapse zones, and abrupt structural changes—are sharply delineated. When examined as time slices or extracted along interpreted layers, the edge detection cube offers a clear and focused visualization of potential geo-hazard zones, supporting more informed well placement and proactive risk mitigation.

Results and Operational Impact

The proposed workflow was tested on multiple 3D seismic datasets within carbonate-rich formations known to exhibit shallow karst development. The performance of the new method was evaluated by comparing its outputs to those of a previously applied approach in Alkhunaini et al. (2022), which involved manual horizon interpretation followed by application of the variance attribute.

Another key benefit was the elimination of interpreter bias during the flattening stage. The previous workflow required the interpretation of at least one seismic horizon to perform structural flattening. This manual step often introduces inconsistencies in areas of low signal-to-noise ratio or complex stratigraphy, making the final attribute output highly dependent on the interpreter's experience and expertise. In contrast, the automated flattening algorithm used in the current study provided consistent results across the entire volume, even in regions lacking clear reflector continuity.

The new edge detection workflow, when applied to the flattened seismic volumes, revealed a 10% increase in geo-hazard detection accuracy compared to the legacy method. This was quantified by correlating anomalies highlighted by the edge detection attribute with known zones of drilling losses and borehole instability recorded in legacy well data (Fig. 3). Many of the karst zones detected by the new method were either missed or only partially imaged in the previous variance-based approach (Fig. 4).

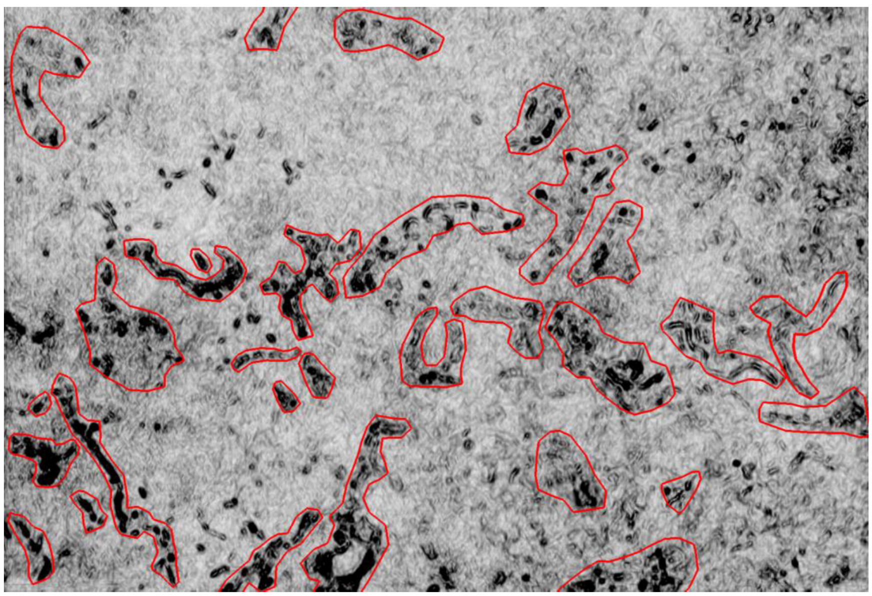

Figure 3—Time slice of an edge detection attribute with an enhanced Sobel-based filter extracted from a 3D seismic volume that was flattened using a horizon-independent flattening algorithm (red polygons used to highlight geo-body anomaly).

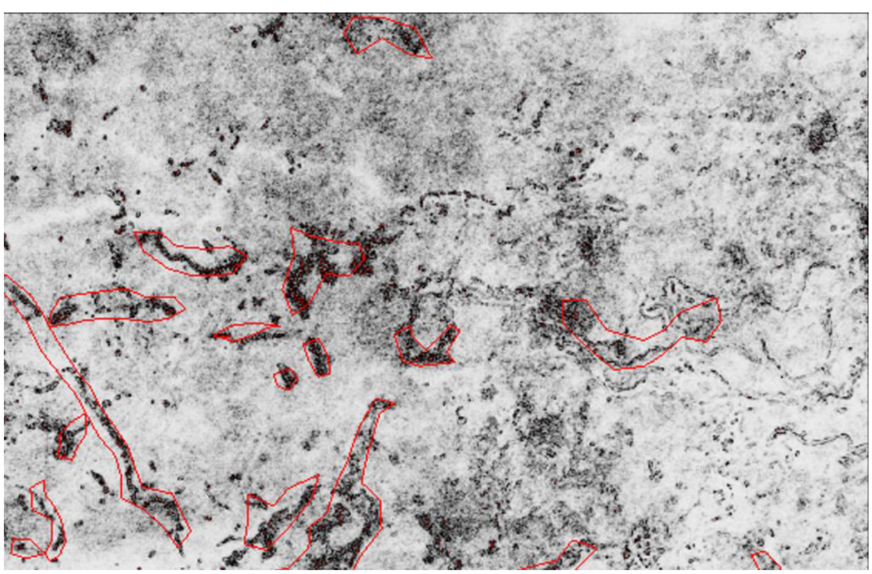

Figure 4—Time slice of a variance attribute extracted from a 3D seismic volume that was flattened on an interpreted horizon (Alkhunaini et al., 2022) (red polygons used to highlight geo-body anomaly).

Time-slice comparisons between the two methods revealed that the Sobel-based edge detection produced higher-resolution and more geologically consistent outlines of karst features (Fig. 5 and Fig. 6). From an operational perspective, the improved detection resolution and automated nature of the workflow enabled faster turnaround of subsurface risk maps. This is especially valuable in an active development setting, where decisions regarding well trajectories and drilling sequences must often be made within tight timeframes.

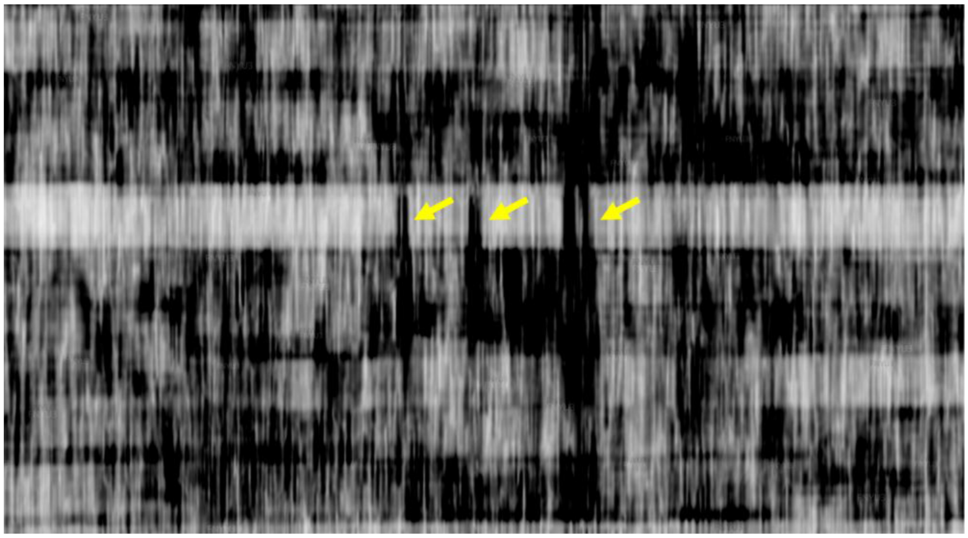

Figure 5—Cross-section of an edge detection attribute with an enhanced Sobel-based filter extracted from a 3D seismic volume that was flattened using a horizon-independent flattening algorithm (Yellow arrows highlighting the geo-hazard anomalies).

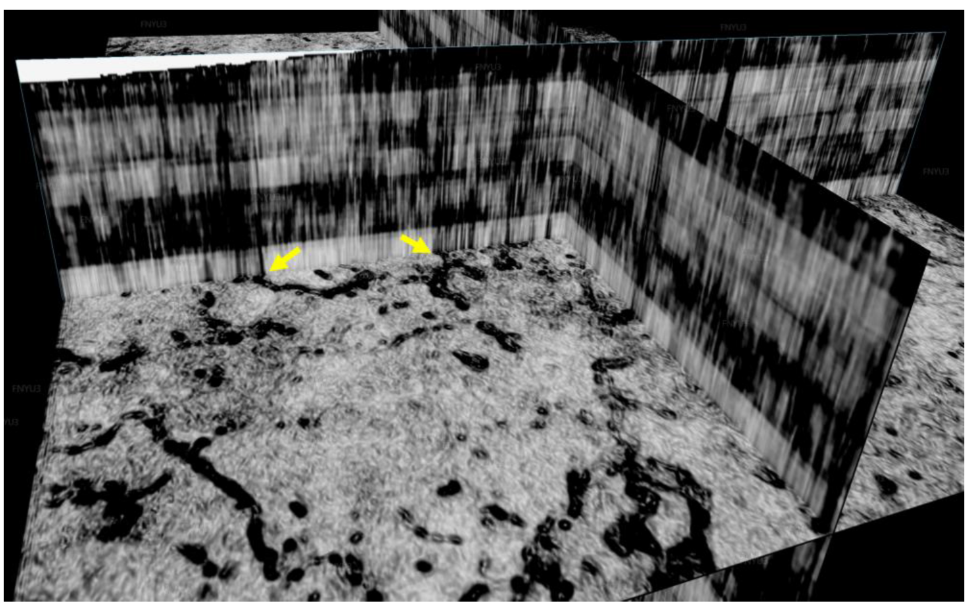

Figure 6—Time-slice and inline/crossline intersections in a 3D view of an edge detection attribute with an enhanced Sobel-based filter extracted from a 3D seismic volume that was flattened using a horizon-independent flattening algorithm (Yellow arrows highlighting the geo-hazard anomalies).

In summary, the results demonstrated that integrating a horizon-free flattening algorithm with a modified Sobel-based edge detection attribute provided a significant step forward in the identification and mapping of shallow carbonate geo-hazards. The workflow not only enhanced detection capability but also improved the consistency, reliability, and interpretive efficiency of the seismic analysis.

Applications, Limitations, and Geomechanical Considerations

Use Cases

The proposed workflow is particularly well-suited to applications in a wide range of resource development projects, where time, accuracy, and risk management are crucial. In shallow carbonate intervals where karsts frequently complicate drilling, having a reliable, scalable, and interpreter-independent hazard detection method provides tangible operational benefits.

One of the most promising use cases is in pre-drill geo-hazard screening. Before committing to a well location, seismic volumes can be processed through this workflow to identify potential collapsed zones, karst voids, and unstable features. This helps engineers design optimized trajectories, select casing points, and prepare for potential drilling fluid losses. The consistency of the horizon-free flattening also makes this method ideal for regional studies, where manual horizon interpretation may not be feasible across large areas of hundreds of square kilometers.

Additionally, the improved edge detection attribute can be integrated with machine learning models as input features for automatic classification of subsurface hazards. In workflows designed to automate seismic interpretation, the quality of input data is crucial. Since this method reduces interpreter bias and highlights subtle discontinuities, it ensures a higher signal-to-noise ratio in the training data, thereby improving the accuracy and generalization of machine learning (ML) models.

Moreover, the technique can be extended to other discontinuity-related use cases such as fault and fracture detection, channel mapping, and stratigraphic boundary identification, particularly when integrated with coherence or amplitude-based attributes. Its utility is not limited to karsts alone—it provides a general framework for highlighting structural and stratigraphic anomalies in any subsurface setting where lateral variation is geologically significant.

Workflow Limitations

Despite its advantages, the workflow has limitations that should be acknowledged. The flattening algorithm, while robust, relies on accurate dip estimation from structure tensors. In areas where the seismic signal is weak, chaotic, or contaminated by acquisition artifacts, dip calculations may be unstable, potentially introducing local misalignments in the flattened volume. While the Vectorized Disorder Algorithm helps mitigate this by down-weighting low-confidence regions, its effectiveness still depends on the quality of the seismic data.

Edge detection, particularly gradient-based methods like Sobel, also presents challenges. Although trace correlation and complex trace enhancements improve accuracy, the filter may still respond to amplitude variations not linked to true geologic features, such as acquisition footprint, processing artifacts, or residual velocity effects. The interpretation of edge maps must be performed with geological context in mind, ideally supported by well data and logs to validate the nature of the detected discontinuities.

Additionally, the current workflow only addresses detection and mapping, and not quantification. While it highlights where hazards are likely to occur, it does not inherently estimate the size, shape, or porosity of karst features—parameters that are equally important for planning drilling or evaluating their reservoir impact. Integrating this method with inversion or amplitude-versus-offset (AVO) analysis may help close this gap in future developments of this method.

Geomechanical Implications

The geomechanical consequences of encountering karstified zones during drilling are severe and well-documented. Karst features are typically intervals of weakness with reduced rock strength, irregular geometry, and high void space. Drilling into such features may lead to partial or total loss of circulation, which in turn compromises hydrostatic pressure and increases the risk of wellbore collapse.

These geo-hazard intervals also complicate cementing and casing operations. The irregularity of voids may prevent proper cement bonding, resulting in poor zonal isolation, gas migration, or annular flow. In the long term, this jeopardizes well integrity, especially during pressure cycling or thermal recovery operations.

By improving the detection and delineation of karst features prior to drilling, the workflow presented in this study directly contributes to safer, more predictable, and more efficient field development.

Conclusion

Shallow geo-hazards, particularly karsts, remain among the most persistent and costly challenges in the development of oil and gas plays. Formed through the long-term dissolution of carbonate rocks, these features pose significant drilling and completion risks, including total circulation losses, borehole instability, poor cement bonding, and compromised long-term well integrity. Their irregular geometries, subsurface unpredictability, and geomechanical complexity make them difficult to detect and mitigate using conventional 3D seismic interpretation techniques.

This study presents a fully automated, interpreter-independent seismic workflow designed to enhance the detection of such subsurface hazards. By integrating a horizon-free flattening algorithm with a geologically optimized, Sobel-based edge detection filter, the workflow addresses key limitations of traditional methods —namely, reliance on manually interpreted horizons and sensitivity to noise in standard seismic attributes. The flattening stage uses structure tensors and eigen decomposition, stabilized by the Vectorized Disorder Algorithm (VDA), to generate stratigraphically coherent seismic volumes even in chaotic or low-reflectivity areas. The edge detection stage applies a modified Sobel-based filter enhanced with trace correlation, complex trace augmentation, and dip guidance, producing clearer and more geologically meaningful delineations of karst-related discontinuities.

Validation against legacy well data demonstrated a 10% improvement in detection accuracy compared to the prior approach, which used interpreted horizons and variance attributes. In addition to increased precision, the workflow offers greater consistency, scalability, and processing efficiency, making it highly suitable for large-scale applications such as pre-drill risk mapping, regional hazard assessment, and integration with machine learning models.

While the method presents clear advantages, it is not without limitations. Dip estimation remains sensitive to seismic quality, and edge detection still requires interpretive validation. Future enhancements could focus on combining this approach with inversion techniques or amplitude-versus-offset (AVO) analysis to more precisely quantify the properties of detected features.

In summary, the integration of horizon-independent flattening and advanced edge detection provides a robust, adaptable, and operationally ready solution for identifying shallow hazards. As the industry increasingly targets geologically complex reservoirs, workflows that combine automation with geological intelligence, such as the one presented here, will be crucial in enhancing subsurface development by improving safety, efficiency, and cost-effectiveness.

Acknowledgment

The authors would like to thank Mohammad R. Mohanna (Saudi Aramco) for his efforts in supporting this project. Moreover, would also like to thank Leticia I. Flores Sanchez and Maria E. Bentosa Gutierrez (Saudi Aramco) for their contributions in documenting and providing the loss of circulation legacy well data used in this study.

References

Fu, Qiang and S. A. Al. Dossary (2021), System and method for randomness measurement in seismic image data using vectorized disorder algorithm, United States Patent US 11,740,374. United States Patent and Trademark Office.

Alkhunaini, Fahad A, Mohanna, Mohammad R., Elisabeth, Francis L. (2022). Integrated approach for de-risking an unconventional tight sand play. *Image 2022*.

Luo, Y., Higgs, W. G. and Kowalik, W. S., (1996). Edge detection and stratigraphic analysis using 3D seismic data: SEG Technical Program Expanded Abstracts.

Brian Russell & Claude Ribordy (2014). New edge detection methods for seismic interpretation, Hampson-Russell, A CGG Company, Calgary, Alberta.

Peter Adetokunbo, Abdullatif A. Al-Shuhail, & Saleh Al-Dossary (2016). 3D Seismic Edge Detection Using Magic Squares and Cubes.